

BSc (Hons) Computer Science
BSc (Hons) Cyber Security
Module Code: QH0305 Module Title: Problem Solving
Assessment Sheet 6

Instructions:

This is one of the eight assessment tasks which will contribute to the overall mark. You will need to complete the tasks as outlined below and then document them in a Word document file. As a minimum, you should provide screenshots of the following:

- Your code
- The output that your code generates

In instances where your code could generate different outputs depending on what values are given, you should provide multiple screenshots of the console screen, showing the different outputs in order to demonstrate that the code works correctly.

Your Word document should have appropriate headings to ensure that each task can easily be identified alongside the rest of your work.

This assessment will focus on sorting algorithm and handling of strings.

You must attempt all tasks on this sheet to achieve a higher grade. For example, if you want to gain marks between 70 - 100, you must complete all other grades first and add them to your portfolio with screenshots.

A zip folder with all Grade codes (in .c or .txt format) must be submitted alongside the portfolio (MS Word file). This is in addition to the code presented as screenshots.

Assessment Task 6: Book Organizer

In this task, you will work with a single array of strings, where each string is a book title. You will progressively display, sort, and manage this catalogue, with complexity increasing across the four grade bands.

To achieve Grade D (Between 40-49):

- Create a string array with 8 book titles in deliberately mixed alphabetical order.
- Use a loop to print on each iteration:
 - Current Title: "<title>"
 - First Title: "<title_at_index_0>"
- No sorting or manipulation is required at this stage.
- Provide screenshots of your code and outputs.

Example output:

Current Title: "North Passage"
First Title: "North Passage"
Current Title: "Echoes of the Atrium"
First Title: "North Passage"
Current Title: "Archive Key"
First Title: "North Passage"

...

To achieve Grade C (Between 50-59):

- Complete all previous steps.
- Modify your loop to perform a single pass that puts the earliest title at index 0 (case-insensitive alphabetical comparison):
 - Compare each title with the one at index 0
 - If a title comes earlier alphabetically, swap it with the one at index 0
- Do not print anything during the execution of the loop.
- Print the playlist before and after the loop.
- If completed successfully, only the first title (alphabetically) should move to the front.

Example Output:

Before: "North Passage", "Echoes of the Atrium", "Archive Key", "Lantern Works", ...

After: "Archive Key", "Echoes of the Atrium", "North Passage", "Lantern Works", ...

To achieve Grade B (Between 60-69):

- Complete all previous steps.
- Implement a nested loop to sort the list in alphabetical order, following similar logic i.e. swapping strings with the first unsorted element of the array on each pass.
- Print the playlist before and after sorting.
- Add comments to your portfolio explaining how the sorting works.

Example output:

Before: "North Passage", "Echoes of the Atrium", "Archive Key", "Lantern Works", ...

After: "Archive Key", "Echoes of the Atrium", "Lantern Works", "North Passage", ...

To achieve Grade A (Between 70-100):

- Complete all previous steps.
- Allow the user to enter a book title to delete from the list.
 - If found, remove the title and shift the remaining titles (so no gap in array).
 - If not found, display an appropriate message.
- Implement a different sorting algorithm (e.g., Bubble Sort).
- Compare the old algorithm with your new one (e.g. Bubble Sort) in terms of logic and performance. This should be explained thoroughly to achieve the highest grade.

Assignment Preparation Guidelines

- All components of the assignment report must be Word-processed (**handwritten text or hand drawn diagrams are not acceptable**), font size must be within the range of 11 point to 14 point including the headings, body text and any texts within diagrams.
- Standard and commonly used fonts such as Times New Roman, Arial or Calibri should be used.
- All figures, graphs and tables must be numbered and labelled with short explanations.
- Material from external sources must be properly acknowledged and cited within the text using the Harvard referencing system.
- All components of the assignment (text, diagrams, code etc.) and all assessment sheets must be submitted in one Word file.
- The report should be logically structured, the core of the report may start by defining the problem / requirements, followed by the proposed solution including a detailed discussion, analysis and evaluation, leading to the implementation and testing stage, finally a conclusion and personal reflection on learning.
- Screenshots without description / discussion are not suitable as they do not express your understanding or support your work adequately.

Submission instructions

- This is a portfolio assignment with eight tasks in total. Each task will be completed and saved in the portfolio. Once the portfolio is completed, it should be submitted on Turnitin. The submission link to Turnitin can be found under the “Assessment Tab” in your module section in the SOL VLE.
- Please note file size limitation might apply. Your report must be under 250MB.
- The source code (in .c or .txt format) for each task should be **zipped** and complete single ZIP file **submitted** to the VLE alongside the report.
- The Assignment Brief can be found under the “Assessment Tab” in your module section in the SOL VLE.
- **Refer to the Assignment Brief** to find the links to Late Submissions, Extenuating Circumstances, Academic Misconduct, Ethics Policy, Grade marking and Guidance for online submission through Solent Online Learning (SOL).